

↑ AFFIX SEAL HERE ↑

CANDIDATE – PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

Signature

TEST CODE **01238010**

MAY/JUNE 2006

FORM TP 2006120

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

PHYSICS

Paper 01 – General Proficiency

75 minutes

29 MAY 2006 (a.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
3. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) newton
- (B) metre
- (C) kilogram
- (D) second

Sample Answer

(A) (B) (C) (D)

The best answer to this item is “metre” so answer space (B) has been shaded.

4. If you want to change your answer, be sure to erase your old answer completely and fill in your new choice.
5. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. You can come back to the harder item later. Your score will be the total number of correct answers.
6. Figures are not necessarily drawn to scale.
7. You may do any rough work in this booklet.
8. The use of non-programmable calculators is allowed.
9. This test consists of 60 items. You will have 75 minutes to answer them.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2005 Caribbean Examinations Council. ®
All rights reserved.

01238010/F2006

1. The base unit of temperature of the S.I. system is the

(A) degree Celsius
(B) degree Fahrenheit
(C) Kelvin
(D) degree centigrade

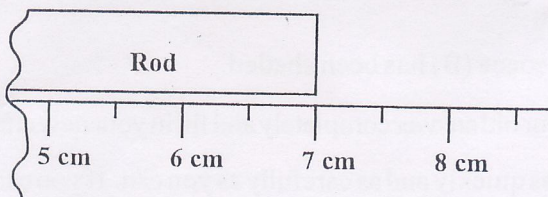
2. If force = mass $\times$ acceleration, the unit of force could be written

(A) $\text{kg}^{-1} \text{m s}^2$
(B) $\text{kg m}^{-1} \text{s}^2$
(C) $\text{kg m}^{-1} \text{s}^2$
(D) kg m s^{-2}

3. Which of the following measurements has three significant figures?

(A) 0.0293 kg
(B) 0.94 A
(C) 5.321 V
(D) 10.42 m

4.



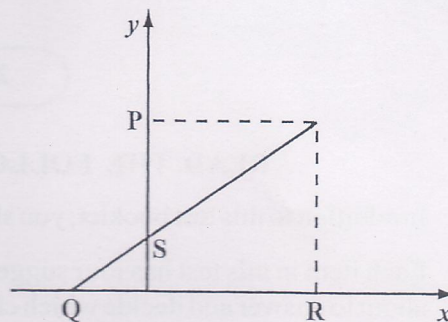
The diagram above shows a rod and a centimetre rule measuring its length. The left end of the rod (not shown) is at the zero mark of the rule. The length of the rod should be written as

(A) 7 cm
(B) 7.0 cm
(C) 7.00 cm
(D) 6.97 cm

5. Which of the following remains unchanged with an increase in temperature?

(A) Density
(B) Mass
(C) Relative Density
(D) Volume

Item 6 refers to the diagram below.



6. When $x = 0$, the value of y is

(A) Q
(B) S
(C) P
(D) P/R

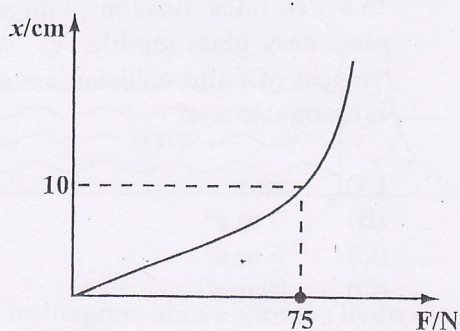
7. Which of the following will be constant, if a constant net force is applied to a body?

(A) Velocity
(B) Kinetic energy
(C) Momentum
(D) Acceleration

8. The newton-metre is the unit used to measure

(A) force
(B) momentum
(C) moment of a force
(D) power

9.



The diagram above shows a simple force (F)/extension (x) graph for a light spring. Which of the following statements would be true?

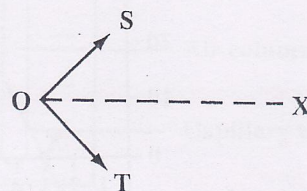
- I. The elastic limit of the spring was exceeded.
- II. The spring obeyed Hooke's law over its entire extension.
- III. The force per unit extension in the elastic region was 7.5 N cm^{-1} .

- (A) I only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

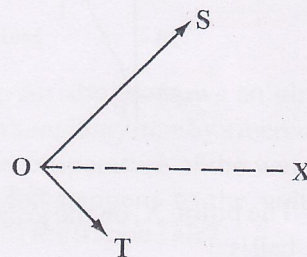
10.

The diagrams below, drawn to scale, represent two forces, S and T , acting at O . In which of the following is the resultant in the direction, OX ?

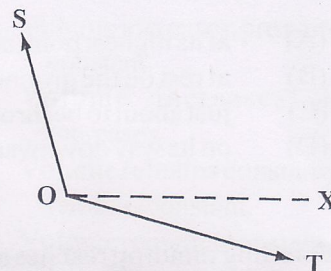
(A)



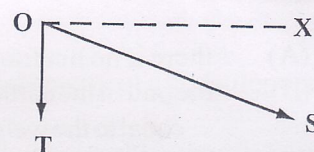
(B)



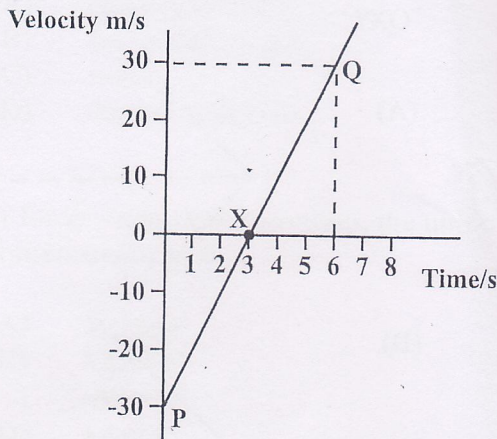
(C)



(D)



Item 11 refers to the following velocity-time graph which shows a ball being thrown vertically upwards from the ground.



11. The point, X, on the graph shows when the ball is

(A) at its highest point above the ground
(B) at rest on the ground
(C) just about to be thrown upwards
(D) on its way downwards

12. A falling raindrop reaches a constant speed when

(A) there is no net force acting on it
(B) the pull of the earth on the raindrop is equal to the weight of the raindrop
(C) the upthrust due to the air is at a minimum
(D) the air surrounding the raindrop becomes saturated with water vapour

13. If an object is moving along a surface with a constant acceleration, the net force acting on the object is

(A) zero
(B) constant
(C) increasing
(D) decreasing

14. A glass marble, X, moving with a speed of 6 m s^{-1} , collides 'head on' with an identical stationary glass marble, Y. What is the velocity of Y after collision, assuming that X is brought to rest?

(A) zero
(B) 3 m s^{-1}
(C) 6 m s^{-1}
(D) 12 m s^{-1}

15. The rate of which of the following will determine the power of a machine?

I. Doing work
II. Converting energy
III. Changing temperature

(A) I only
(B) III only
(C) I and II only
(D) I, II and III

16. Which of the following implements is/are designed to take advantage of a large moment provided by a relatively small force?

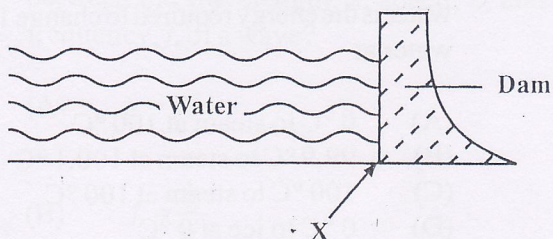
I. Clawhammer
II. Crowbar
III. Pair of tweezers
IV. Pair of wire cutters

(A) III only
(B) I and IV only
(C) I, II and III only
(D) I, II and IV only

17. The upthrust experienced by a solid immersed in a liquid is equal to the

(A) mass of the solid
(B) volume of the liquid displaced
(C) weight of the liquid displaced
(D) density of the liquid

18.



The diagram above shows a hydroelectric dam. The pressure on the dam at point X depends on the

- (A) depth of the water
- (B) volume of water held by the dam
- (C) mass of water held back by the dam
- (D) length of the reservoir

19.

A bubble of gas rises to the surface of a soft drink. This is because the

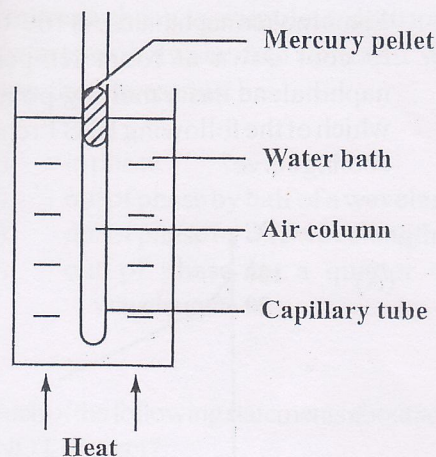
- (A) upthrust on the bubble is greater than the weight of the bubble
- (B) upthrust on the bubble is greater than the weight of water it displaces
- (C) weight of water displaced by the bubble is less than the weight of the bubble
- (D) density of the gas is greater than the density of the drink

20.

Which of the following is the MOST suitable range for a clinical thermometer?

- (A) 0°C to 44°C
- (B) 10°C to 110°C
- (C) 35°C to 100°C
- (D) 35°C to 44°C

21.



The diagram above shows an air column trapped in a capillary tube by a mercury pellet. When the temperature of the water bath is raised, what happens to the volume and pressure of the trapped air?

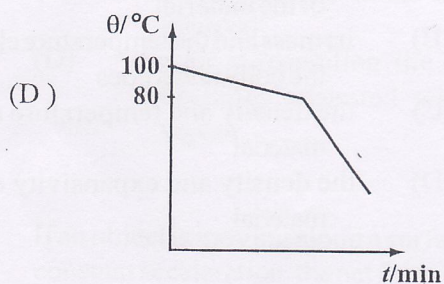
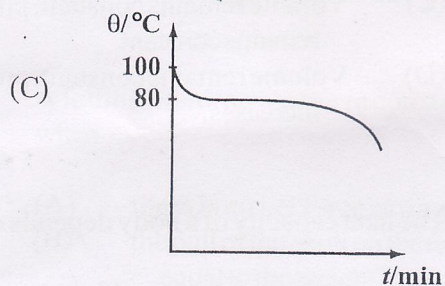
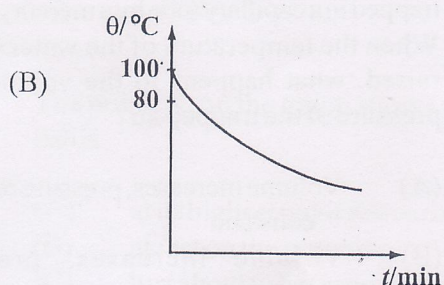
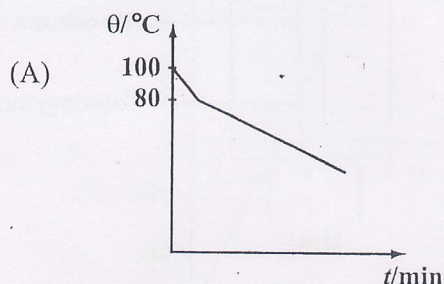
- (A) Volume increases, pressure remains constant.
- (B) Volume increases, pressure increases.
- (C) Volume remains constant, pressure remains constant.
- (D) Volume remains constant, pressure increases.

22.

The heat capacity of a body depends on

- (A) its mass and the specific heat capacity of the material
- (B) its mass and the temperature change the body undergoes
- (C) the density and temperature of the material
- (D) the density and expansivity of the material

23. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C , which of the following BEST represents the cooling curve?



24. The specific latent heat of vapourization of water is the energy required to change 1 kg of water at

- (A) 0°C to steam at 100°C
- (B) 99.9°C to steam at 100.1°C
- (C) 100°C to steam at 100°C
- (D) 0°C to ice at 0°C

25. An electric kettle full of water is plugged into the electrical mains supply. The process by which heat travels through the water is

- (A) electrification
- (B) evaporation
- (C) convection
- (D) radiation

26. In which of the following is conduction the main method of energy transfer?

- (A) Food heated in a microwave oven
- (B) Energy transferred from the sun to earth
- (C) Food being cooked on a barbecue
- (D) Food being cooked in a pot on an electric stove

27. Which of the following phenomena are exhibited by light waves?

- I. Refraction
- II. Diffraction
- III. Interference

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

28. Which of the following is the correct relation between the wavelength, λ , speed, v , and frequency, f , of a wave?

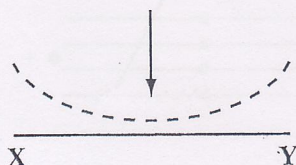
(A) $\lambda = fv$

(B) $f = \frac{\lambda}{v}$

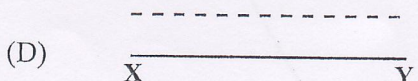
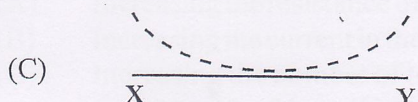
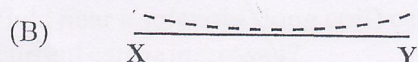
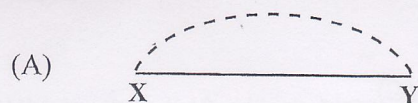
(C) $f = \frac{v}{\lambda}$

(D) $\lambda = \frac{f}{v}$

29.



The figure above shows a wavefront about to reach a reflecting surface XY. Which of the following would represent the wavefront after reflection?



30. Whenever there is complete destructive interference between two coherent wave trains, the waves must be

- (A) in phase
(B) out of phase by half of a wavelength
(C) out of phase by one wavelength
(D) out of phase by a quarter of a wavelength

31. Which of the following statements about sound is NOT correct?

- (A) Sound is transmitted as transverse waves.
(B) Sound may be produced by vibrating systems.
(C) Sound does not travel through a vacuum.
(D) Sound travels more slowly than light.

32. In which of the following are the electromagnetic waves written in order of increasing frequency, starting with the lowest?

- (A) Xrays, ultraviolet, infrared, radio
(B) Xrays, infrared, ultraviolet, radio
(C) Radio, xrays, infrared, ultraviolet
(D) Radio, infrared, ultraviolet, xrays

33. Which of the following is the reason why the diffraction of light is NOT usually observed?

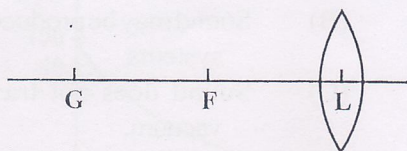
- (A) Its wavelength is too small.
(B) Its frequency is too high.
(C) Its speed is too high.
(D) Its wavelength is too large.

34. Upon which of the following would the position of an image formed by a plane mirror depend?

- I. Distance of the observer from the mirror
- II. Distance of the object from the mirror
- III. Angle at which the image is viewed

- (A) I only
- (B) II only
- (C) II and III only
- (D) I, II and III

35.



The above diagram represents a thin converging lens, L, with a principal axis GL. FL is the focal length of the lens and $GF = FL$. Where should an object be placed so as to produce a virtual magnified image?

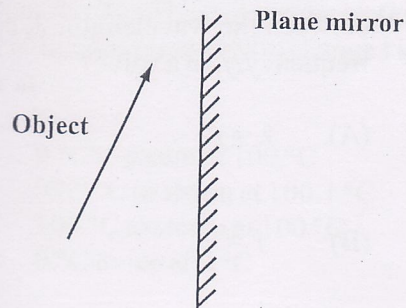
- (A) At G
- (B) To the left of G
- (C) Between F and G
- (D) Between L and F

36. Which of the following can produce a diminished virtual image of a real object?

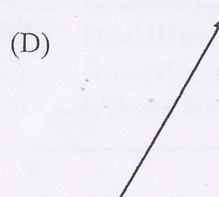
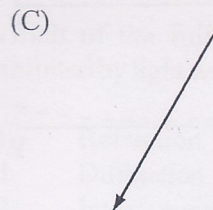
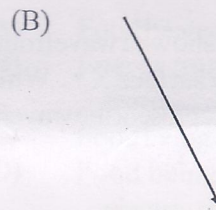
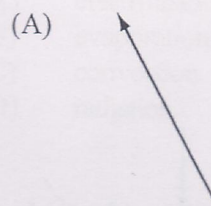
- I. A plane mirror
- II. A diverging lens
- III. A converging lens

- (A) I only
- (B) II only
- (C) II and III only
- (D) I, II and III

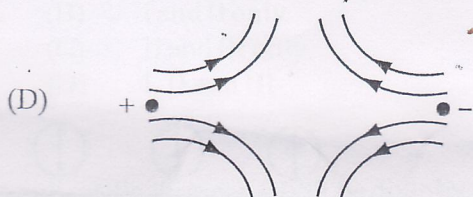
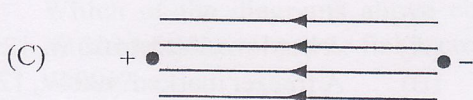
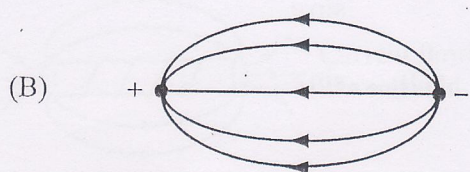
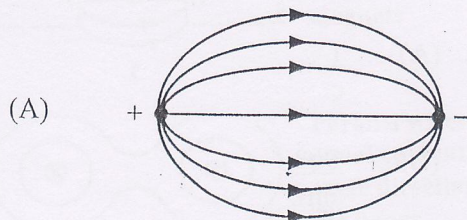
37.



The diagram above represents an object placed in front of a plane mirror. Which of the following BEST represents the image produced by the plane mirror?



38. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?



39. Which of the following is NOT one of the ways in which the strength of the magnetic field near a solenoid (long coil) carrying a current can be increased?

- (A) Increasing the resistance of the coil
- (B) Increasing the current in the coil
- (C) Increasing the number of turns per unit length of the coil
- (D) Placing a soft iron core inside the coil

40. Which of the following relationships between electrical quantities is correct?

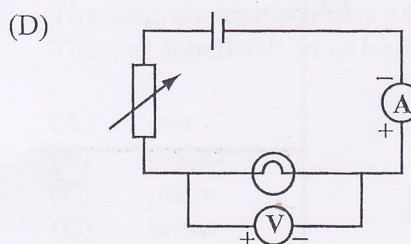
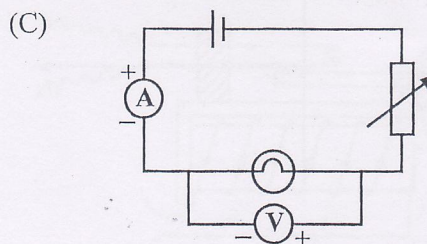
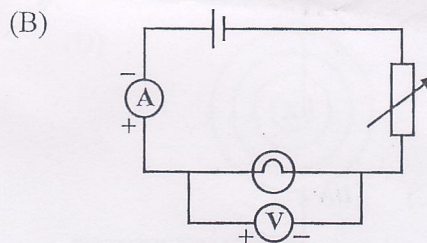
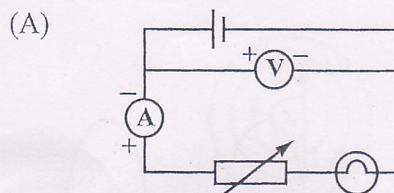
(A) $V = P I$

(B) $R = V I$

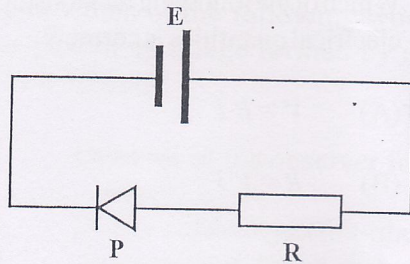
(C) $Q = \frac{E}{V}$

(D) $E = V I$

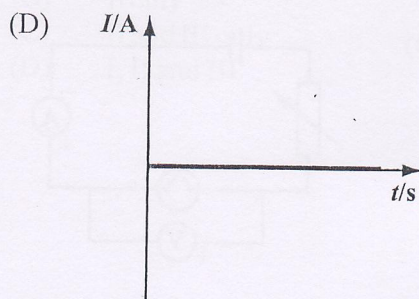
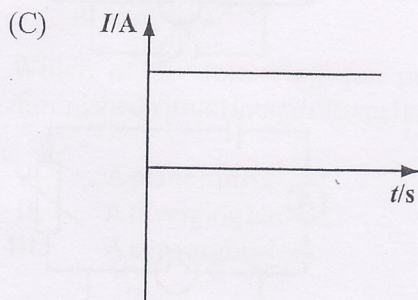
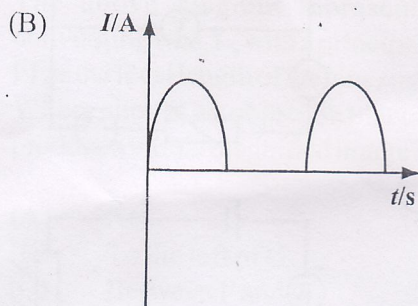
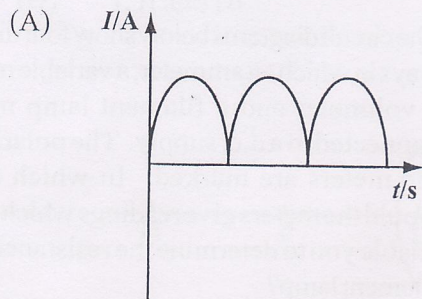
41. The circuit diagrams below show four different ways in which an ammeter, a variable resistor, a voltmeter and a filament lamp may be connected to a d.c. supply. The polarities of the meters are marked. In which circuit would the meters give readings which would enable you to determine the resistance of the filament lamp?



42.



The figure above shows a cell E, a diode P and a resistor, R, connected in series. The current through R is BEST illustrated by which of the following graphs?



43.

The refractive index of a transparent medium with a critical angle, c , for light travelling from the medium to air is

(A) $\frac{1}{c}$

(B) $\frac{90^\circ}{\sin c}$

(C) $\frac{\sin 90^\circ}{\sin c}$

(D) $\sin c$

44.

Which of the following appliances would require the fuse with the largest rating?

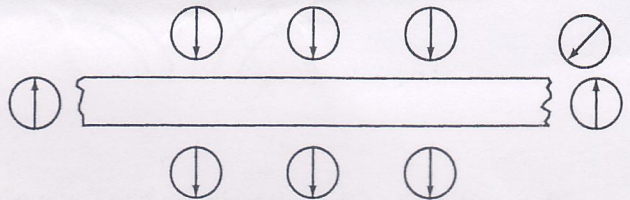
(A) A kettle marked 1 500 W, 120 V

(B) A freezer marked 400 W, 120 V

(C) An iron marked 1 000 W, 240 V

(D) A stove marked 10 000 W, 240 V

45.



In the diagram above, a piece of rubber magnet from a refrigerator is surrounded by plotting compasses. The rubber magnet has

(A) poles at the ends

(B) poles at the centre

(C) poles at the sides

(D) no poles

46.

Steel is NOT suitable for use in an electromagnet because it

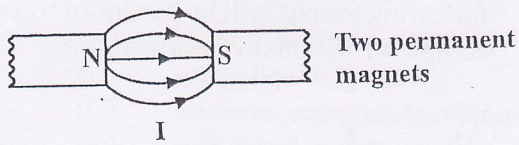
(A) is too difficult to magnetise

(B) retains magnetism too well

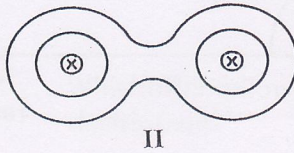
(C) loses its magnetism too easily

(D) contains too many domains

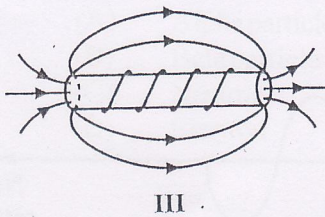
47.



Two permanent magnets



Parallel wires carrying equal currents in the same direction into the paper



Current flowing in a solenoid

Which of the diagrams above correctly illustrate(s) the magnetic fields created as shown?

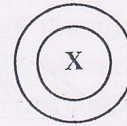
- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

48. Rectification can BEST be done by using a

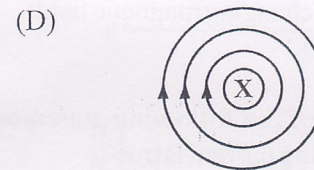
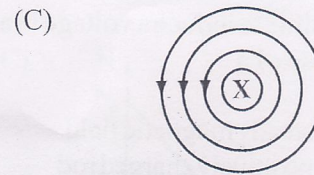
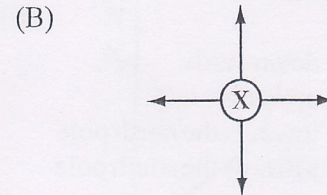
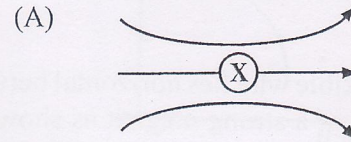
- (A) transformer
- (B) capacitor
- (C) transistor
- (D) diode

49.

The diagram below represents a straight wire carrying a current into the plane of the paper.

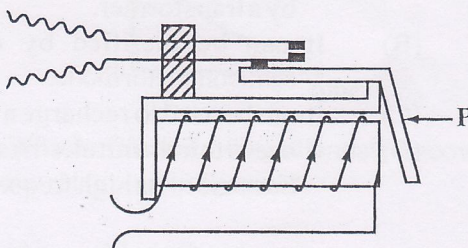


Which of the following diagrams BEST represents the magnetic field around the wire?



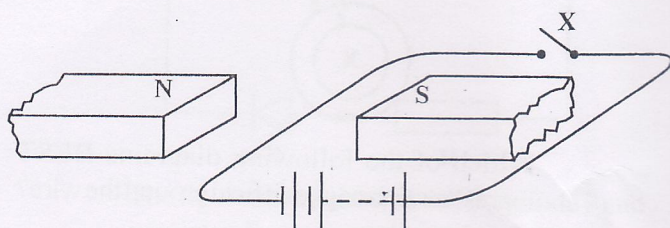
50.

The diagram above shows a typical relay. The part, labelled P, must be made of



- (A) iron
- (B) copper
- (C) plastic
- (D) brass

51.



A flexible wire lies horizontal between the poles of a strong magnet as shown in the diagram above. When the switch, X, is closed the wire moves

- (A) downwards
- (B) upwards
- (C) towards the north pole
- (D) towards the south pole

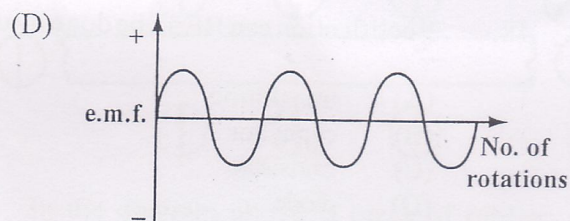
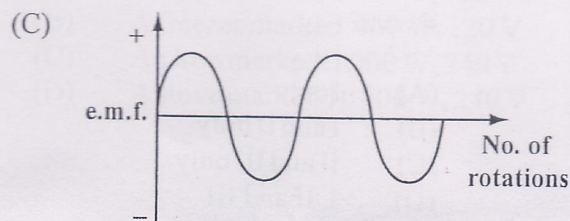
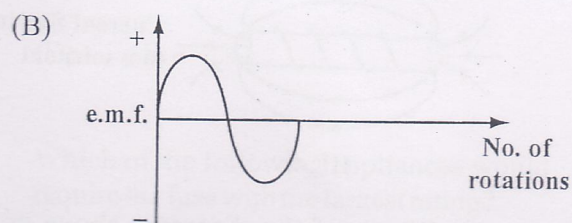
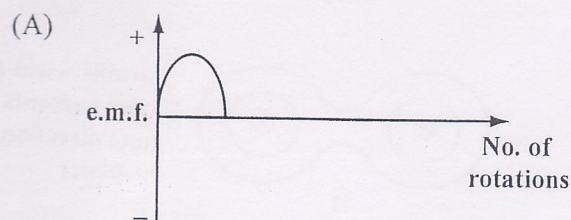
52. It is possible to induce a voltage in a wire coil by the use of a

- (A) steady magnetic field
- (B) positively charged rod
- (C) negatively charged rod
- (D) changing magnetic field

53. Which of the following statements about alternating current is true?

- (A) It can be changed into direct current by a transformer.
- (B) It can be rectified by using a semiconductor diode.
- (C) It can be used to recharge a battery.
- (D) It is used to transmit electrical energy because of its high frequency.

54. For a simple a.c. generator, which of the following graphs will be produced by two complete rotations of the armature?



55. No practical transformer is 100% efficient. Which of the following cause energy losses?

- I. The resistance of the coils of wire
- II. Magnetization and demagnetization of the core
- III. Eddy currents in the core

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

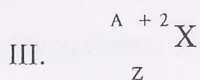
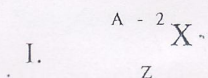
56. The basic constituents of all matter are

- (A) protons, neutrons, electrons
- (B) solids, liquids, gases
- (C) elements, compounds, mixtures
- (D) α , β , γ radiations

57. Which of the following is NOT affected by an electric field?

- (A) Alpha particle
- (B) Beta particle
- (C) Neutron
- (D) Electron

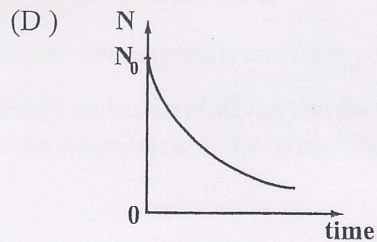
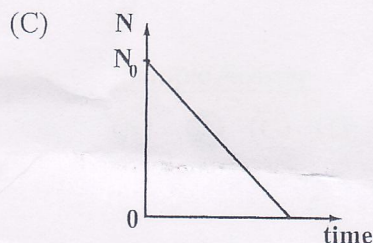
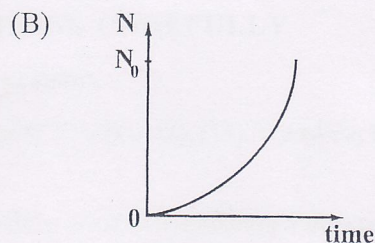
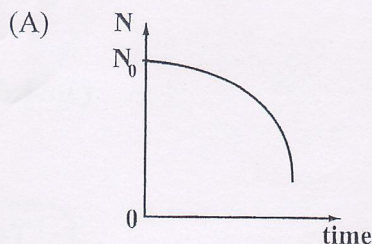
58. Which of the following would be possible symbols for an isotope of nuclide represented by A_ZX ?



- (A) I only
- (B) II only
- (C) III only
- (D) I and III only

59. The number of radioactive nuclei present in a sample at time $t = 0$ is N_0 .

Which of the following graphs BEST represents the variation with time of the number, N , of undecayed nuclei present?



60. Which of the following scientists discovered the relationship $E = mc^2$?

- (A) Marie Curie
- (B) Isaac Newton
- (C) Albert Einstein
- (D) J. J. Thompson